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JEE MAIN PHYSICS DPP: MOTION UNDER GRAVITY

Topic: Free Fall, Vertical Projectiles & Tower Release Problems | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** A ball is thrown vertically upwards with velocity u . The total time of flight for the ball to return to the projection level is:
- $\frac{u}{g}$
 - $\frac{2u}{g}$
 - $\frac{u^2}{2g}$
 - $\frac{\sqrt{2}u}{g}$
- Q2.** A particle is projected vertically upwards. If t_1 and t_2 are the times taken by the particle to reach the same height h on its ascent and descent respectively, the initial projection speed u is:
- $\frac{1}{2}g(t_2 - t_1)$
 - $g(t_1 + t_2)$
 - $\frac{1}{2}g(t_1 + t_2)$
 - $\sqrt{gt_1 t_2}$
- Q3.** A body dropped from the top of a tower of height H traverses a distance of $\frac{5H}{9}$ in the last second of its fall. The total height H of the tower is: (Take $g = 10 \text{ m/s}^2$)
- 80 m
 - 125 m
 - 20 m
 - 45 m
- Q4.** A stone is dropped into a well of depth $d = 20 \text{ m}$. The splash of sound is heard after t seconds. If the speed of sound is 340 m/s and $g = 10 \text{ m/s}^2$, the total time t is:
- 2.06 s
 - 2.00 s
 - 2.12 s
 - 1.94 s
- Q5.** Drops of water fall at regular intervals from the roof of a building of height 16 m. The first drop reaches the ground at the instant the fifth drop starts falling. The height of the third drop from the ground at that instant is:
- 4 m
 - 12 m
 - 9 m
 - 7 m
- Q6.** A balloon is ascending vertically with a constant acceleration of 2 m/s^2 . Two seconds after release from the ground, a stone is dropped from it. The maximum height reached by the stone above the ground is: (Take $g = 10 \text{ m/s}^2$)
- 4.0 m
 - 4.4 m
 - 4.8 m
 - 5.2 m
- Q7.** A stone is thrown vertically upwards with speed u_1 from the top of a tower of height H and reaches the ground in time t_1 . Another stone thrown vertically downwards with the same speed u_1 reaches the ground in time t_2 . If a third stone is simply dropped from rest from the same height, the time it takes to reach the ground is:
- $\frac{t_1 + t_2}{2}$
 - $\frac{t_1 - t_2}{2}$
 - $\frac{t_1 t_2}{t_1 + t_2}$
 - $\sqrt{t_1 t_2}$
- Q8.** In the previous question, the height H of the tower in terms of t_1 and t_2 is:
- $\frac{1}{2}gt_1 t_2$
 - $gt_1 t_2$
 - $\frac{1}{4}gt_1 t_2$
 - $\frac{1}{2}g(t_1 + t_2)^2$
- Q9.** A particle is thrown vertically upwards. The ratio of the distance traversed in the 1st second to that in the 2nd second of its upward journey is 5 : 3. The maximum height reached is: (Take $g = 10 \text{ m/s}^2$)
- 40 m
 - 45 m
 - 50 m
 - 80 m
- Q10.** A rocket is fired vertically upwards with a net acceleration $a = 10 \text{ m/s}^2$. Its fuel is exhausted after 1 minute (60 s). The total time elapsed from blast-off until the rocket returns to the ground is: (Take $g = 10 \text{ m/s}^2$, $\sqrt{2} \approx 1.41$)

- (a) 120 s
(b) 180 s
(c) $60(2 + \sqrt{2}) \text{ s} \approx 205 \text{ s}$
(d) 240 s
- Q11.** A ball A is dropped from the top of a 100 m tower and at the same time another ball B is projected vertically upwards from the base of the tower with speed 50 m/s. The two balls will meet after a time of: (Take $g = 10 \text{ m/s}^2$)
- (a) 1.0 s
(b) 3.0 s
(c) 4.0 s
(d) 2.0 s
- Q12.** In Q11, the height above the ground at which the two balls meet is:
- (a) 80 m
(b) 20 m
(c) 60 m
(d) 75 m
- Q13.** A body is released from a great height H and falls freely. The ratio of the times taken to fall through consecutive equal distances h ($t_1 : t_2 : t_3 : \dots$) is:
- (a) $1 : 3 : 5 : 7 : \dots$
(b) $1 : (\sqrt{2} - 1) : (\sqrt{3} - \sqrt{2}) : \dots$
(c) $1 : \sqrt{2} : \sqrt{3} : \dots$
(d) $1 : 2 : 3 : 4 : \dots$
- Q14.** A juggler maintains 5 balls in motion, tossing each ball to a vertical height of 20 m. The time interval between successive tosses must be: (Take $g = 10 \text{ m/s}^2$)
- (a) 0.5 s
(b) 1.0 s
(c) 0.8 s
(d) 0.4 s
- Q15.** A stone is dropped from a height h . Simultaneously, another stone is thrown up from the ground with such a velocity that it can reach a maximum height of $4h$. The time after which the two stones pass each other is:
- (a) $\sqrt{\frac{h}{2g}}$
(b) $\sqrt{\frac{2h}{g}}$
(c) $\sqrt{\frac{h}{4g}}$
(d) $\sqrt{\frac{h}{8g}}$
- Q16.** A particle is projected vertically upwards with speed u . The average speed of the particle in the time interval from $t = 0$ to $t = \frac{2u}{g}$ is:
- (a) $\frac{u}{2}$
(b) u
(c) $\frac{u}{4}$
(d) 0
- Q17.** A body dropped from the top of a tower covers a distance x in the first second and a distance y in the second second. The value of $\frac{y}{x}$ is:
- (a) 2
(b) 3
(c) 4
(d) 5
- Q18.** A particle is released from rest from a height. The distance travelled by the particle in the 3rd second of its motion is: (Take $g = 10 \text{ m/s}^2$)
- (a) 45 m
(b) 15 m
(c) 25 m
(d) 35 m
- Q19.** A ball is thrown vertically upwards with speed u . At what height above the projection point is the kinetic energy of the ball equal to three times its potential energy (taking ground as reference PE = 0)?
- (a) $\frac{u^2}{2g}$
(b) $\frac{u^2}{4g}$
(c) $\frac{3u^2}{8g}$
(d) $\frac{u^2}{8g}$
- Q20.** A stone dropped from the top of a vertical cliff traverses 44.1 m in the last second before hitting the water below. The height of the cliff is: (Take $g = 9.8 \text{ m/s}^2$)
- (a) 122.5 m
(b) 100.0 m
(c) 144.0 m
(d) 78.4 m

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

 Setting net vertical displacement $y(T) = 0$:

$$uT - \frac{1}{2}gT^2 = 0 \implies T\left(u - \frac{1}{2}gT\right) = 0$$

 Since $T > 0$, total time of flight $T = \frac{2u}{g}$.

Sol 2. Correct Option: (c)

 Equation of motion for height h :

$$h = ut - \frac{1}{2}gt^2 \implies \frac{1}{2}gt^2 - ut + h = 0$$

 t_1 and t_2 are roots of this quadratic equation. Sum of roots:

$$t_1 + t_2 = \frac{u}{\frac{1}{2}g} = \frac{2u}{g} \implies u = \frac{1}{2}g(t_1 + t_2)$$

Sol 3. Correct Option: (d)

 Total time to fall H is $T = \sqrt{2H/g}$.

Distance in last second:

$$s_T = \frac{g}{2}(2T - 1) = \frac{5H}{9}$$

 Since $H = \frac{1}{2}gT^2 \implies \frac{g}{2}(2T - 1) = \frac{5}{9}\left(\frac{1}{2}gT^2\right)$.

$$9(2T - 1) = 5T^2 \implies 5T^2 - 18T + 9 = 0$$

$$(5T - 3)(T - 3) = 0 \implies T = 3 \text{ s } (T > 1 \text{ s})$$

$$H = \frac{1}{2}g(3)^2 = \frac{1}{2}(10)(9) = 45 \text{ m}$$

Sol 4. Correct Option: (a)

• Time for stone to fall:

$$t_1 = \sqrt{\frac{2d}{g}} = \sqrt{\frac{2(20)}{10}} = 2 \text{ s}$$

• Time for sound to travel up:

$$t_2 = \frac{d}{v_{\text{sound}}} = \frac{20}{340} = 0.0588 \text{ s} \approx 0.06 \text{ s}$$

 Total time $t = t_1 + t_2 = 2 + 0.06 = 2.06 \text{ s}$.

Sol 5. Correct Option: (b)

Total time for 1st drop to reach ground:

$$T = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2(16)}{g}} = \frac{4\sqrt{2}}{\sqrt{g}}$$

Number of equal time intervals between 5 drops is 4.

$$\text{Interval } \Delta t = \frac{T}{4}.$$

When 5th drop starts, 3rd drop has fallen for time

$$t = 2\Delta t = \frac{T}{2}.$$

Distance fallen by 3rd drop:

$$y_3 = \frac{1}{2}g\left(\frac{T}{2}\right)^2 = \frac{1}{4}\left(\frac{1}{2}gT^2\right) = \frac{16}{4} = 4 \text{ m}$$

 Height from ground = $16 - 4 = 12 \text{ m}$.

Sol 6. Correct Option: (c)

 At $t = 2 \text{ s}$ of ascent with $a = 2 \text{ m/s}^2$:

 Height $h_0 = \frac{1}{2}at^2 = \frac{1}{2}(2)(2)^2 = 4 \text{ m}$.

 Velocity $v_0 = at = 2 \times 2 = 4 \text{ m/s}$ (upward).

Additional height reached under gravity:

$$h_1 = \frac{v_0^2}{2g} = \frac{4^2}{2(10)} = \frac{16}{20} = 0.8 \text{ m}$$

 Maximum height = $h_0 + h_1 = 4 + 0.8 = 4.8 \text{ m}$.

Sol 7. Correct Option: (d)

Taking downward direction as positive from tower top:

 • Upward projection: $-u_1t_1 + \frac{1}{2}gt_1^2 = H \implies u_1 = \frac{1}{2}gt_1 - \frac{H}{t_1}$.

 • Downward projection: $u_1t_2 + \frac{1}{2}gt_2^2 = H \implies u_1 = \frac{H}{t_2} - \frac{1}{2}gt_2$.

Equating gives:

$$\frac{H}{t_1} + \frac{H}{t_2} = \frac{1}{2}g(t_1 + t_2)$$

$$H\left(\frac{t_1 + t_2}{t_1t_2}\right) = \frac{1}{2}g(t_1 + t_2)$$

$$H = \frac{1}{2}gt_1t_2$$

 For free fall: $H = \frac{1}{2}gt^2 \implies \frac{1}{2}gt^2 = \frac{1}{2}gt_1t_2 \implies t = \sqrt{t_1t_2}$.

Sol 8. Correct Option: (a)

From the previous worked solution:

$$H = \frac{1}{2}gt_1t_2$$

Sol 9. Correct Option: (b)

$$s_1 = u - \frac{1}{2}g(1) = u - 5$$

$$s_2 = u - \frac{1}{2}g(3) = u - 15$$

$$\text{Given } \frac{u-5}{u-15} = \frac{5}{3}:$$

$$3(u-5) = 5(u-15) \implies 3u-15 = 5u-75$$

$$2u = 60 \implies u = 30 \text{ m/s}$$

Maximum height:

$$H_{\max} = \frac{u^2}{2g} = \frac{30^2}{2(10)} = \frac{900}{20} = 45 \text{ m}$$

Sol 10. Correct Option: (c)

- Phase 1 (Engine on, $t_1 = 60$ s):

$$v_1 = at_1 = 10 \times 60 = 600 \text{ m/s.}$$

$$h_1 = \frac{1}{2}at_1^2 = \frac{1}{2}(10)(60)^2 = 18000 \text{ m.}$$

- Phase 2 (Ascending under gravity to peak):

$$t_2 = \frac{v_1}{g} = \frac{600}{10} = 60 \text{ s.}$$

$$h_2 = \frac{v_1^2}{2g} = \frac{360000}{20} = 18000 \text{ m.}$$

$$\text{Total height } H = 18000 + 18000 = 36000 \text{ m.}$$

- Phase 3 (Free fall from peak):

$$t_3 = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2(36000)}{10}} = \sqrt{7200} = 60\sqrt{2} \text{ s.}$$

Total time:

$$T = t_1 + t_2 + t_3 = 60 + 60 + 60\sqrt{2} = 60(2 + \sqrt{2}) \text{ s}$$

Sol 11. Correct Option: (d)

In relative frame of ball B :

$$u_{\text{rel}} = u_A - u_B = 0 - (-50) = 50 \text{ m/s.}$$

$$a_{\text{rel}} = (-g) - (-g) = 0.$$

Relative distance $S = 100 \text{ m.}$

$$t = \frac{S}{u_{\text{rel}}} = \frac{100}{50} = 2.0 \text{ s}$$

Sol 12. Correct Option: (a)

For ball B projected upward:

$$y_B(2) = 50(2) - \frac{1}{2}(10)(2)^2 = 100 - 20 = 80 \text{ m}$$

Sol 13. Correct Option: (b)

$$\text{Time to fall through distance } nh: T_n = \sqrt{\frac{2nh}{g}} =$$

$$\sqrt{n} \cdot \sqrt{\frac{2h}{g}}.$$

$$\bullet t_1 = T_1 = \sqrt{1}\tau$$

$$\bullet t_2 = T_2 - T_1 = (\sqrt{2} - 1)\tau$$

$$\bullet t_3 = T_3 - T_2 = (\sqrt{3} - \sqrt{2})\tau$$

$$\text{Ratio: } 1 : (\sqrt{2} - 1) : (\sqrt{3} - \sqrt{2}) : \dots$$

Sol 14. Correct Option: (c)

Total time in air for each ball:

$$T = 2\sqrt{\frac{2h}{g}} = 2\sqrt{\frac{2(20)}{10}} = 2(2) = 4.0 \text{ s}$$

With 5 balls in continuous circulation, interval:

$$\Delta t = \frac{T}{5} = \frac{4.0}{5} = 0.8 \text{ s}$$

Sol 15. Correct Option: (d)

Projection speed of second stone:

$$H_{\max} = \frac{u^2}{2g} = 4h \implies u = \sqrt{8gh}$$

Meeting time with relative acceleration = 0:

$$t = \frac{h}{u_{\text{rel}}} = \frac{h}{\sqrt{8gh}} = \sqrt{\frac{h^2}{8gh}} = \sqrt{\frac{h}{8g}}$$

Sol 16. Correct Option: (a)

Total distance travelled in one round trip:

$$\text{Distance} = 2H_{\max} = 2\left(\frac{u^2}{2g}\right) = \frac{u^2}{g}$$

$$\text{Total time } T = \frac{2u}{g}.$$

Average speed:

$$\text{Speed}_{\text{avg}} = \frac{\text{Distance}}{T} = \frac{u^2/g}{2u/g} = \frac{u}{2}$$

Sol 17. Correct Option: (b)

$$x = s_1 = \frac{1}{2}g(1)^2 = 5 \text{ m.}$$

$$y = s_2 = \frac{1}{2}g(2)^2 - \frac{1}{2}g(1)^2 = 20 - 5 = 15 \text{ m.}$$

$$\text{Ratio: } \frac{y}{x} = \frac{15}{5} = 3.$$

Sol 18. Correct Option: (c)

Distance in n th second from rest:

$$s_n = \frac{g}{2}(2n-1)$$

For $n = 3$:

$$s_3 = \frac{10}{2}(2(3)-1) = 5(5) = 25 \text{ m}$$

Sol 19. Correct Option: (d)

$$\text{Total mechanical energy: } E = \frac{1}{2}mu^2.$$

$$\text{Given } K = 3U \implies E = K + U = 4U = 4(mgh).$$

$$\frac{1}{2}mu^2 = 4mgh \implies h = \frac{u^2}{8g}$$

Sol 20. Correct Option: (a)

$$s_T = \frac{g}{2}(2T-1) = 44.1.$$

$$\frac{9.8}{2}(2T-1) = 44.1 \implies 4.9(2T-1) = 44.1$$

$$2T-1 = 9 \implies 2T = 10 \implies T = 5 \text{ s}$$

Height of cliff:

$$H = \frac{1}{2}gT^2 = \frac{1}{2}(9.8)(5)^2 = 4.9 \times 25 = 122.5 \text{ m}$$

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